



MINING DEALER BEST PRACTICE

Swing Motor Sensors Position Change on HMS 6060AC

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

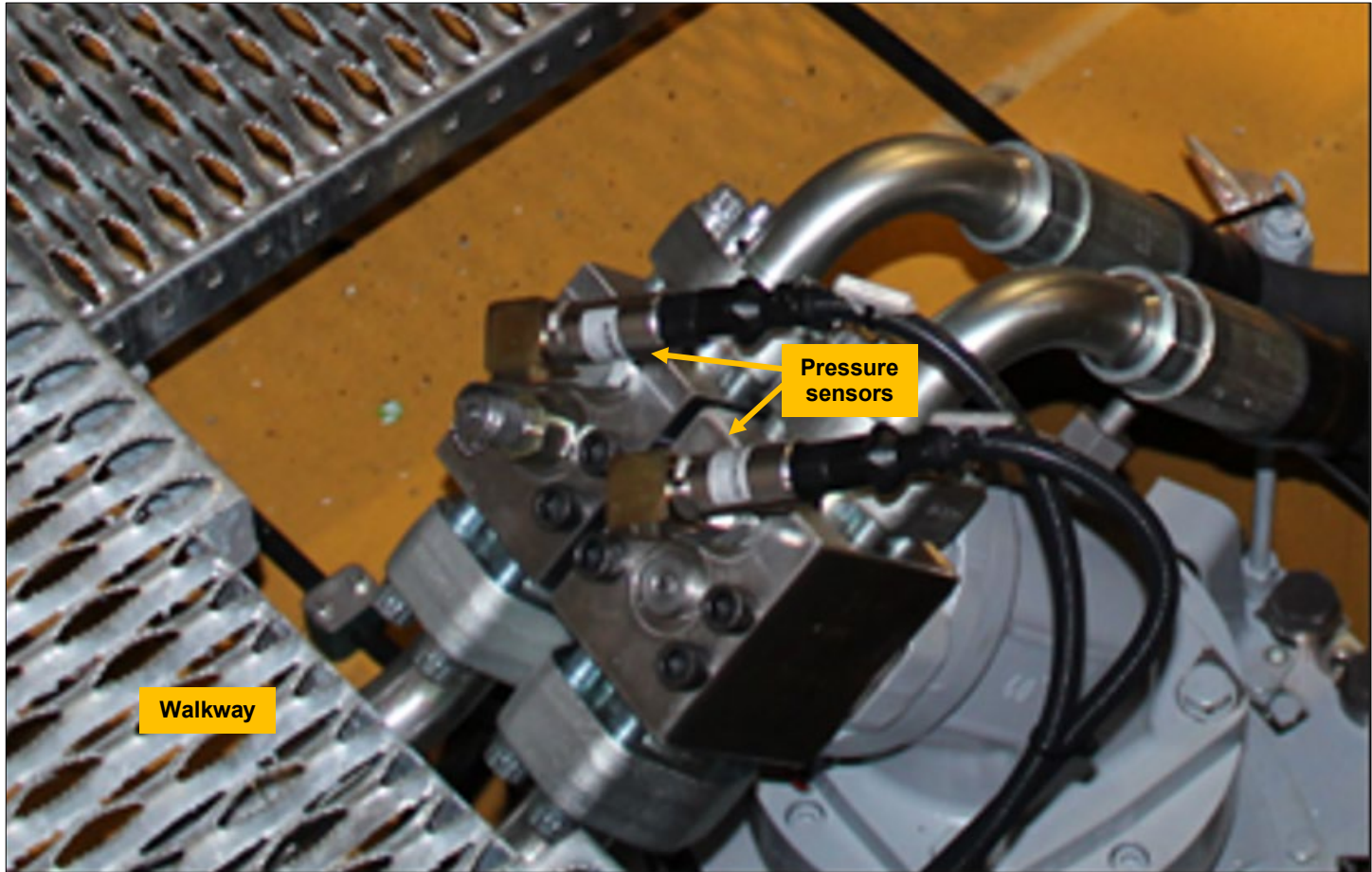
Swing Motor Sensors Position Change on HMS 6060AC	1
1.0 Introduction.....	2
2.0 Best Practice Description	2
3.0 Implementation Steps.....	4
4.0 Benefits	4
5.0 Resources Required.....	5
6.0 Supporting Attachments / References	5
7.0 Related Best Practices	5
8.0 Acknowledgements	5

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices.” Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

February 2023
1.04-230202-1

1.0 Introduction

The two pressure sensors (PN 534-4587) and the temperature sensor (PN 534-4592) of HMS 6060AC swing system are located on the rear swing motors closest to the walkway, as shown on the picture 1. Therefore, due to the position of these sensors, they can be easily damaged by people walking on the walkway or at service / maintenance work in this area.

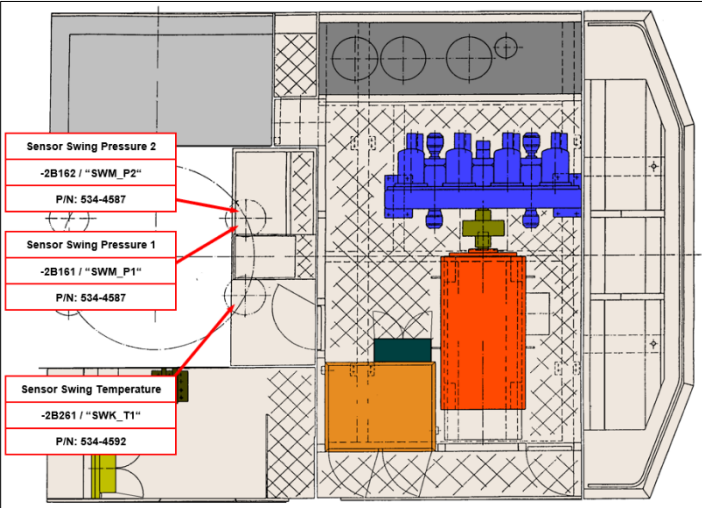


Picture 1: Pressure sensors PN 534-4587 applied at rear swing motors, near the walkway.

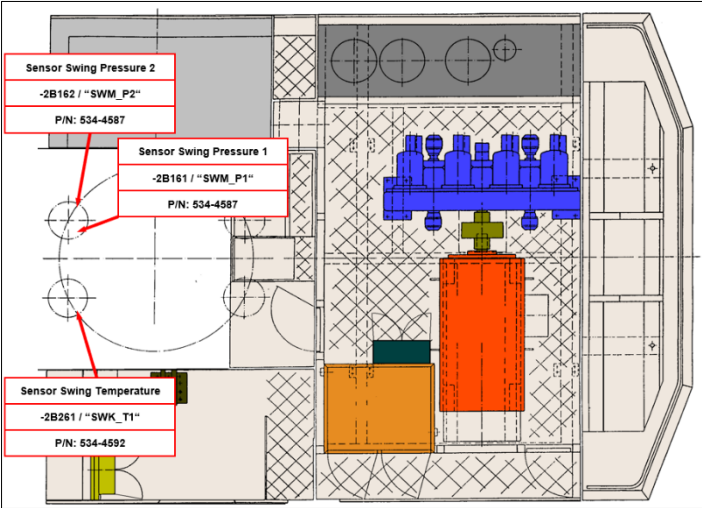
2.0 Best Practice Description

To avoid damages of these sensors, they should be removed from the rear swing motors to the front swing motors. Pictures 2 to 4 show details about this modification. In this area, the sensors are protected from almost all external damage, and this saves costs and downtime.

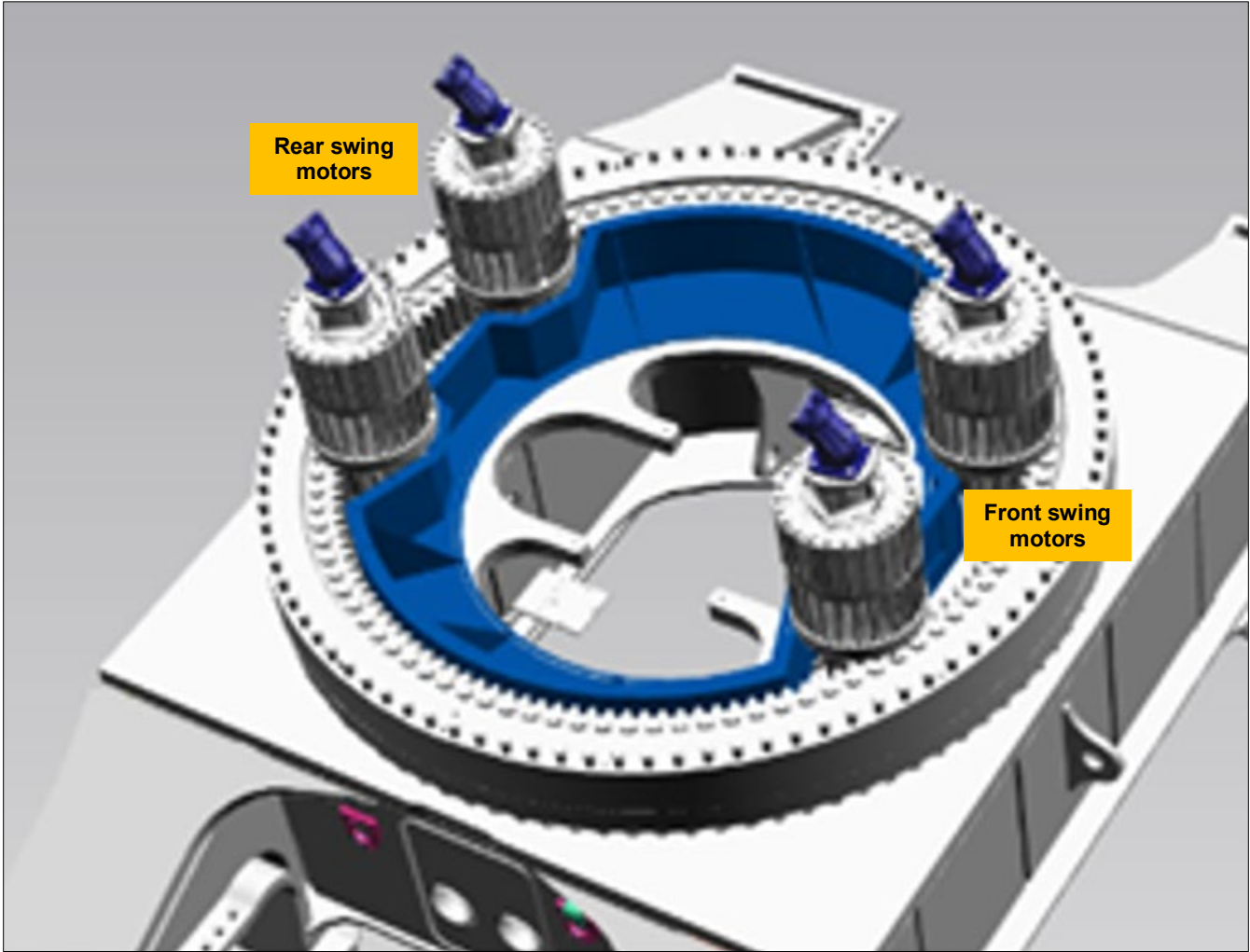
THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Swing Motor Sensors Position Change on HMS 6060AC	DATE 2/2/2023	CHG NO 00	NUMBER 1.04-230202-1



Picture 2: Original position of sensors at swing motors - rear of machine.



Picture 3: Position of sensors at swing motors after modification - front of machine.



Picture 4: Rear and front swing motor's location at machine model 6060AC.

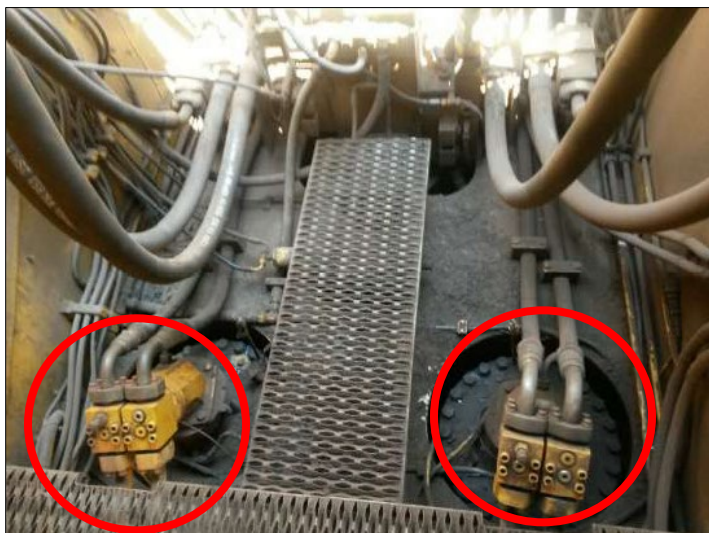
THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Swing Motor Sensors Position Change on HMS 6060AC	DATE	CHG NO	NUMBER
	2/2/2023	00	1.04-230202-1

3.0 Implementation Steps

Note: Parts list in the "5.0 Resources Required"

1. Stop and shutdown the machine. Lock the safety device preventing the machine from switching on, apply a warning sign.
2. Remove the sensors installed on the rear swing motors and install the plugs and seals (items #1 and #2 of parts list) at same location.
3. At front swing motors, install the manifolds (items #3 and #8 of parts list) one on each motor and install all the flanges and adapters (items #4 to #7 of parts list).
4. Install the pressure and temperature sensors at the manifolds installed on the front swing motors;
5. Develop 3 electrical harness using the wire and conduit (items #11 and #12 of parts list). The connectors of these harness are Deutsch type (items #9 and #10 of parts list). Each harness has 4 meters and a female connector in terminal and male connector in other;
6. Connect these harnesses at original harness and at the sensors.

The pictures 5 and 6 show details of all swing motors after installation.



Picture 5: Rear swing motors after installation, without sensors.



Picture 6: Front swing motors after installation, with sensors installed on the manifolds.

4.0 Benefits

1. Maintenance: Eliminate failures by people walking near the sensors.
2. Productivity: Contribute to improve the Mean Time Before Failure (MTBF) of the machine.
3. Safety: Easy access.
4. Commercial: All items involved have CAT part numbers.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Swing Motor Sensors Position Change on HMS 6060AC	DATE	CHG NO	NUMBER
	2/2/2023	00	1.04-230202-1

5.0 Resources Required

The table 1 below show all the items used for this improvement.

ITEM	PART NUMBER	DESCRIPTION	QUANTITY
1	367-5991	Plug	04 ea
2	155-9644	Seal	04 ea
3	507-4242	Manifold	02 ea
4	451-4733	Seal	12 ea
5	438-9272	Bolt	16 ea
6	487-2026	Adapter	04 ea
7	484-8870	Flange	08 ea
8	534-1181	Bolt	16 ea
9	155-2260	Kit connector plug 3 pin	03 ea
10	9W-0844	Pin	12 ea
11	Local supply	Electrical wire 1.5 mm	24 meters
12	Local supply	Electrical conduit	12 meters

Table 1: Items used in this improvement.

Total parts cost US\$: 2,518.16

Labor: 8 hour – 2 technicians

6.0 Supporting Attachments / References

BI017741 - Parts catalogue 6060AC FS

SEHS9615-03 - Servicing Deutsch HD and DT style connectors

Parts list - see attached

7.0 Related Best Practices

N/A

8.0 Acknowledgements

This Best Practice was written by:

João Marinho

joao.marinho@sotreq.com.br

This Best Practice was written and implanted by:

Alexsander Faria

alexsander.faria@sotreq.com.br

João Marinho

joao.marinho@sotreq.com.br

Fabio Leonardo

fabio.leonardo@sotreq.com.br

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Swing Motor Sensors Position Change on HMS 6060AC	DATE 2/2/2023	CHG NO 00	NUMBER 1.04-230202-1

Best Practice Champion

Fellipe Viana

fellipe.viana@sotreq.com.br

Special thanks in development to:

Antonio Frieiro

antonio.frieiro@sotreq.com.br

Ary Rizel

ary.rizel@sotreq.com.br

Carlos Dias

carlos.dias@sotreq.com.br

Cirley Ribeiro

cirley.ribeiro@sotreq.com.br

Cassio Pereira

Cassio.Pereira@sotreq.com.br

Daniel Cunha

daniel.cunha@sotreq.com.br

Davy Dutra

davy.dutra@sotreq.com.br

Douglas Reis

douglas.reis@sotreq.com.br

Fabio Xavier

Fabio.Xavier@sotreq.com.br

Heder Coelho

Heder.Coelho@sotreq.com.br

Janair Xavier

janair.xavier@sotreq.com.br

Jud Oliveira

jud.oliveira@sotreq.com.br

Michael Silva

michael.freire@sotreq.com.br

Paulo Pirani

paulo.pirani@sotreq.com.br

Rodolpho Wachsmuth

rodolpho.wachsmuth@sotreq.com.br

Ronan Duarte

ronan.duarte@sotreq.com.br

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Swing Motor Sensors Position Change on HMS 6060AC	DATE 2/2/2023	CHG NO 00	NUMBER 1.04-230202-1

Thiago RezendeThiago.Rezende@sotreq.com.br**Ygor Ceolin**ygor.ceolin@sotreq.com.br**Yuri Oliveira**ygor.ceolin@sotreq.com.br

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Swing Motor Sensors Position Change on HMS
6060AC

DATE
2/2/2023

CHG
NO
00

NUMBER
1.04-230202-1